No.	Status	Date
HIS-2006-0412-DS-3-RV-101	Issued	2006-02-24

UT Thickness Data Sheet - 3-RV-101

Scope and Responsibilities

This data sheet provides ultrasonic thickness measurements for line 3-RV-101, located within the ATU-100. The objective is to record current wall thickness and evaluate the asset integrity of the regenerator overhead piping as part of the 10-year full-unit survey.

Kevin Doucette, acting as lead UT technician (SNT-TC-1A UT Level II) for Halward Inspection Services Ltd., is responsible for the execution of the measurements and the recording of the data presented herein.

Technical Basis and Procedures

Thickness measurements were performed in accordance with the site NDE procedure QSCP-NDT-001 Rev 0 and the Halward Inspection Services Ltd. procedure HIS-UT-002 Rev 1. All ultrasonic testing was conducted following the guidelines established in ASTM E797.

The data collected for this line supports the findings and conclusions detailed in the comprehensive report HIS-2006-0412.

Component Design Data

The subject line, 3-RV-101, is constructed from A106 Gr.B with a nominal outside diameter of 88.9 mm and a nominal wall thickness of 7.62 mm. The design pressure is 345 kPa(g) and the design temperature is 121 °C.

The minimum allowable thickness (t-min) for this component is 3.56 mm. Any reading below this threshold is subject to the site non-conformance reporting process.

Integrity Assessment

As a regenerator overhead line in the ATU-100, 3-RV-101 is designated as Class 1. This classification requires thickness and external visual inspections at a maximum interval of 5 years.

Remaining life and corrosion rates are calculated based on the current reading relative to the t-min of 3.56 mm. The next inspection date is determined by half the remaining life or the Class 1 maximum interval.

Thickness readings, measured 2006-02-24

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2001-03-02	6.82	6.59	0.053	0.046	>50	2011-02-24	
6-SG-101-02	7.11	3.56	2001-03-02	6.71	6.51	0.054	0.040	>50	2011-02-24	
6-SG-101-03	7.11	3.56	2001-03-02	6.74	6.42	0.066	0.064	43.2	2011-02-24	
6-SG-101-04	7.11	3.56	2001-03-02	6.73	6.28	0.081	0.090	30.1	2011-02-24	
6-SG-101-05	7.11	3.56	2001-03-02	6.68	6.22	0.092	0.092	28.8	2011-02-24	
6-SG-101-06	7.11	3.56	2001-03-02	6.87	6.49	0.058	0.076	38.4	2011-02-24	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2001-03-02	6.72	6.26	0.088	0.092	29.2	2011-02-24	
6-SG-102-02	7.11	3.56	2001-03-02	6.72	6.25	0.086	0.094	28.5	2011-02-24	
6-SG-102-03	7.11	3.56	2001-03-02	6.69	6.26	0.078	0.086	31.3	2011-02-24	
6-SG-102-04	7.11	3.56	2001-03-02	6.64	6.22	0.077	0.084	31.6	2011-02-24	
6-SG-102-05	7.11	3.56	2001-03-02	6.93	6.72	0.041	0.042	>50	2011-02-24	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	1996-03-08	8.20	7.89	0.031	0.031	>50	2016-02-24	
8-TG-101-02	8.18	4.06	1996-03-08	8.20	7.75	0.045	0.045	>50	2016-02-24	
8-TG-101-03	8.18	4.06	1996-03-08	8.23	7.79	0.044	0.044	>50	2016-02-24	
8-TG-101-04	8.18	4.06	1996-03-08	8.10	7.76	0.034	0.034	>50	2016-02-24	
8-TG-101-05	8.18	4.06	1996-03-08	8.22	7.95	0.027	0.027	>50	2016-02-24	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	1996-03-08	5.99	5.51	0.048	0.048	48.4	2016-02-24	
4-LA-101-02	6.02	3.18	1996-03-08	6.01	5.76	0.025	0.025	>50	2016-02-24	
4-LA-101-03	6.02	3.18	1996-03-08	5.96	5.76	0.020	0.020	>50	2016-02-24	
4-LA-101-04	6.02	3.18	1996-03-08	6.07	5.78	0.029	0.029	>50	2016-02-24	
4-LA-101-05	6.02	3.18	1996-03-08	6.01	5.81	0.020	0.020	>50	2016-02-24	
4-LA-101-06	6.02	3.18	1996-03-08	6.06	5.64	0.042	0.042	>50	2016-02-24	
4-LA-101-07	6.02	3.18	1996-03-08	5.99	5.57	0.042	0.042	>50	2016-02-24	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	1996-03-08	6.02	5.63	0.039	0.039	>50	2016-02-24	
4-LA-102-02	6.02	3.18	1996-03-08	6.07	5.67	0.040	0.040	>50	2016-02-24	
4-LA-102-03	6.02	3.18	1996-03-08	6.07	5.55	0.052	0.052	45.4	2016-02-24	
4-LA-102-04	6.02	3.18	1996-03-08	5.99	5.34	0.065	0.065	33.1	2016-02-24	
4-LA-102-05	6.02	3.18	1996-03-08	6.00	5.62	0.038	0.038	>50	2016-02-24	
4-LA-102-06	6.02	3.18	1996-03-08	6.00	5.83	0.017	0.017	>50	2016-02-24	
4-LA-102-07	6.02	3.18	1996-03-08	6.00	5.63	0.037	0.037	>50	2016-02-24	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	1996-03-08	7.02	6.75	0.027	0.027	>50	2016-02-24	
6-LA-103-02	7.11	3.56	1996-03-08	7.19	6.87	0.032	0.032	>50	2016-02-24	
6-LA-103-03	7.11	3.56	1996-03-08	7.08	6.65	0.043	0.043	>50	2016-02-24	
6-LA-103-04	7.11	3.56	1996-03-08	7.17	6.52	0.065	0.065	45.4	2016-02-24	
6-LA-103-05	7.11	3.56	1996-03-08	7.19	6.49	0.070	0.070	41.7	2016-02-24	
6-LA-103-06	7.11	3.56	1996-03-08	7.16	6.42	0.074	0.074	38.5	2016-02-24	
6-LA-103-07	7.11	3.56	1996-03-08	7.18	6.76	0.042	0.042	>50	2016-02-24	
6-LA-103-08	7.11	3.56	1996-03-08	7.04	6.79	0.025	0.025	>50	2016-02-24	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2001-03-02	8.39	8.27	0.025	0.024	>50	2016-02-24	
4-RA-101-02	8.56	3.81	2001-03-02	8.45	8.27	0.027	0.036	>50	2016-02-24	
4-RA-101-03	8.56	3.81	2001-03-02	8.42	8.10	0.051	0.064	>50	2016-02-24	
4-RA-101-04	8.56	3.81	2001-03-02	8.37	8.34	0.029	0.006	>50	2016-02-24	
4-RA-101-05	8.56	3.81	2001-03-02	8.42	8.29	0.031	0.026	>50	2016-02-24	
4-RA-101-07	8.56	3.81	2001-03-02	8.51	8.30	0.030	0.042	>50	2016-02-24	
4-RA-101-08	8.56	3.81	2001-03-02	8.30	8.07	0.061	0.046	>50	2016-02-24	
4-RA-101-09	8.56	3.81	2001-03-02	8.43	8.43	0.014	0.000	>50	2016-02-24	
4-RA-101-10	8.56	3.81	2001-03-02	8.48	8.27	0.040	0.042	>50	2016-02-24	
4-RA-101-11	8.56	3.81	2001-03-02	8.39	8.14	0.054	0.050	>50	2016-02-24	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2001-03-02	8.48	8.41	0.013	0.014	>50	2016-02-24	
4-RA-102-02	8.56	3.81	2001-03-02	8.50	8.15	0.042	0.070	>50	2016-02-24	
4-RA-102-03	8.56	3.81	2001-03-02	8.37	8.27	0.029	0.020	>50	2016-02-24	
4-RA-102-04	8.56	3.81	2001-03-02	8.35	8.24	0.030	0.022	>50	2016-02-24	
4-RA-102-05	8.56	3.81	2001-03-02	8.36	8.19	0.036	0.034	>50	2016-02-24	
4-RA-102-06	8.56	3.81	2001-03-02	8.38	8.15	0.036	0.046	>50	2016-02-24	
4-RA-102-07	8.56	3.81	2001-03-02	8.49	8.29	0.029	0.040	>50	2016-02-24	
4-RA-102-08	8.56	3.81	2001-03-02	8.36	8.05	0.044	0.062	>50	2016-02-24	
4-RA-102-09	8.56	3.81	2001-03-02	8.35	8.18	0.037	0.034	>50	2016-02-24	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2001-03-02	10.82	10.76	0.024	0.012	>50	2016-02-24	
6-RA-103-02	10.97	4.57	2001-03-02	10.69	10.50	0.059	0.038	>50	2016-02-24	
6-RA-103-03	10.97	4.57	2001-03-02	10.84	10.67	0.032	0.034	>50	2016-02-24	
6-RA-103-04	10.97	4.57	2001-03-02	10.59	10.40	0.058	0.038	>50	2016-02-24	
6-RA-103-05	10.97	4.57	2001-03-02	10.61	10.34	0.061	0.054	>50	2016-02-24	
6-RA-103-06	10.97	4.57	2001-03-02	10.86	10.74	0.027	0.024	>50	2016-02-24	
6-RA-103-07	10.97	4.57	2001-03-02	10.79	10.66	0.033	0.026	>50	2016-02-24	
6-RA-103-08	10.97	4.57	2001-03-02	10.83	10.58	0.039	0.050	>50	2016-02-24	
6-RA-103-09	10.97	4.57	2001-03-02	10.85	10.67	0.034	0.036	>50	2016-02-24	
6-RA-103-10	10.97	4.57	2001-03-02	10.78	10.27	0.075	0.102	>50	2016-02-24	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2001-03-02	7.26	6.98	0.061	0.056	>50	2011-02-24	
3-RV-101-02	7.62	3.56	2001-03-02	7.31	6.93	0.064	0.076	44.2	2011-02-24	
3-RV-101-03	7.62	3.56	2001-03-02	7.15	6.57	0.095	0.116	25.9	2011-02-24	
3-RV-101-04	7.62	3.56	2001-03-02	7.18	6.65	0.094	0.106	29.1	2011-02-24	
3-RV-101-05	7.62	3.56	2001-03-02	7.25	7.01	0.066	0.048	>50	2011-02-24	
3-RV-101-06	7.62	3.56	2001-03-02	7.20	6.74	0.085	0.092	34.4	2011-02-24	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2001-03-02	5.29	4.85	0.068	0.088	23.3	2011-02-24	
2-RV-102-02	5.54	2.79	2001-03-02	5.12	4.84	0.075	0.056	27.2	2011-02-24	
2-RV-102-03	5.54	2.79	2001-03-02	5.24	5.14	0.034	0.020	>50	2011-02-24	
2-RV-102-04	5.54	2.79	2001-03-02	5.33	5.07	0.051	0.052	43.7	2011-02-24	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	1996-03-08	5.48	5.48	0.000	0.000	>50	2016-02-24	
3-RF-101-02	5.49	2.41	1996-03-08	5.45	5.37	0.008	0.008	>50	2016-02-24	
3-RF-101-03	5.49	2.41	1996-03-08	5.45	5.45	0.000	0.000	>50	2016-02-24	
3-RF-101-04	5.49	2.41	1996-03-08	5.46	5.33	0.013	0.013	>50	2016-02-24	
3-RF-101-05	5.49	2.41	1996-03-08	5.45	5.40	0.005	0.005	>50	2016-02-24	
3-RF-101-06	5.49	2.41	1996-03-08	5.52	5.40	0.012	0.012	>50	2016-02-24	
3-RF-101-07	5.49	2.41	1996-03-08	5.42	5.35	0.007	0.007	>50	2016-02-24	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	1996-03-08	3.92	3.85	0.007	0.007	>50	2016-02-24	
2-RF-102-02	3.91	2.16	1996-03-08	3.88	3.88	0.000	0.000	>50	2016-02-24	
2-RF-102-03	3.91	2.16	1996-03-08	3.93	3.80	0.013	0.013	>50	2016-02-24	
2-RF-102-04	3.91	2.16	1996-03-08	3.87	3.81	0.006	0.006	>50	2016-02-24	
2-RF-102-07	3.91	2.16	1996-03-08	3.96	3.85	0.011	0.011	>50	2016-02-24	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	1996-03-08	6.07	5.75	0.032	0.032	>50	2016-02-24	
4-LA-104-02	6.02	3.18	1996-03-08	6.03	5.56	0.047	0.047	>50	2016-02-24	
4-LA-104-03	6.02	3.18	1996-03-08	5.96	5.74	0.022	0.022	>50	2016-02-24	
4-LA-104-04	6.02	3.18	1996-03-08	6.06	5.86	0.020	0.020	>50	2016-02-24	
4-LA-104-05	6.02	3.18	1996-03-08	6.04	5.51	0.053	0.053	43.8	2016-02-24	
4-LA-104-06	6.02	3.18	1996-03-08	6.10	5.83	0.027	0.027	>50	2016-02-24	
4-LA-104-07	6.02	3.18	1996-03-08	6.01	5.62	0.039	0.039	>50	2016-02-24	
4-LA-104-08	6.02	3.18	1996-03-08	5.94	5.50	0.044	0.044	>50	2016-02-24	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2001-03-02	8.40	8.16	0.035	0.048	>50	2016-02-24	
4-RA-104-02	8.56	3.81	2001-03-02	8.41	8.23	0.032	0.036	>50	2016-02-24	
4-RA-104-03	8.56	3.81	2001-03-02	8.46	8.30	0.024	0.032	>50	2016-02-24	
4-RA-104-04	8.56	3.81	2001-03-02	8.35	8.21	0.038	0.028	>50	2016-02-24	
4-RA-104-05	8.56	3.81	2001-03-02	8.29	8.14	0.046	0.030	>50	2016-02-24	
4-RA-104-06	8.56	3.81	2001-03-02	8.39	8.09	0.050	0.060	>50	2016-02-24	
4-RA-104-07	8.56	3.81	2001-03-02	8.29	7.97	0.062	0.064	>50	2016-02-24	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	1996-03-08	3.93	3.64	0.029	0.029	46.4	2016-02-24	
2-ST-101-02	3.91	2.29	1996-03-08	3.90	3.67	0.023	0.023	>50	2016-02-24	
2-ST-101-03	3.91	2.29	1996-03-08	3.94	3.82	0.012	0.012	>50	2016-02-24	
2-ST-101-04	3.91	2.29	1996-03-08	3.82	3.77	0.005	0.005	>50	2016-02-24	
2-ST-101-05	3.91	2.29	1996-03-08	3.91	3.72	0.019	0.019	>50	2016-02-24	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	1996-03-08	5.42	5.13	0.029	0.029	>50	2016-02-24	
3-CW-101-02	5.49	2.79	1996-03-08	5.41	5.20	0.021	0.021	>50	2016-02-24	
3-CW-101-03	5.49	2.79	1996-03-08	5.53	5.08	0.045	0.045	>50	2016-02-24	
3-CW-101-04	5.49	2.79	1996-03-08	5.46	5.38	0.008	0.008	>50	2016-02-24	
3-CW-101-05	5.49	2.79	1996-03-08	5.48	5.33	0.015	0.015	>50	2016-02-24	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	1996-03-08	5.50	5.20	0.030	0.030	>50	2016-02-24	
3-CW-102-02	5.49	2.79	1996-03-08	5.51	5.02	0.049	0.049	45.4	2016-02-24	
3-CW-102-03	5.49	2.79	1996-03-08	5.53	5.07	0.046	0.046	49.4	2016-02-24	
3-CW-102-04	5.49	2.79	1996-03-08	5.51	5.03	0.048	0.048	46.5	2016-02-24	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2006-02-24	_____